



Foundations of Agentic AI for Retail

Concepts, Technologies, and Architectures for Autonomous Retail Systems

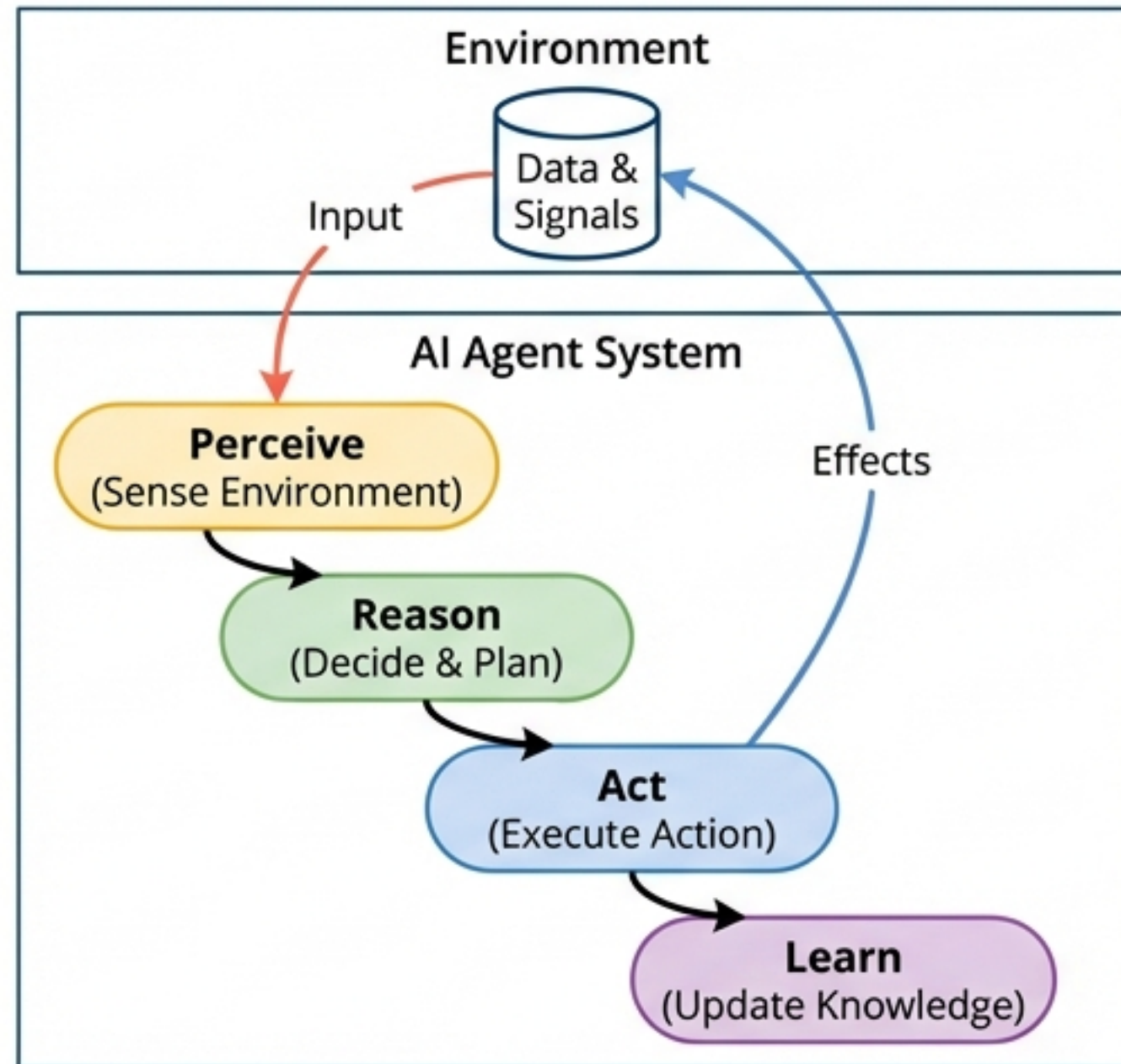
Based on the 2025 work by Dr. Fatih Nayebi | Prepared by Graph4VA

The Evolution: From Reactive Automation to Proactive Autonomy



Key Insight: Agentic AI does not just analyze; it autonomously executes tasks, designs workflows, and uses tools to achieve business goals.

The Core Loop: Perceive, Reason, Act, Learn



Perceive:

Ingests real-time signals from the environment (Sales data, IoT sensors, Web analytics).

Reason: Plans and decides using cognitive engines (often LLMs). Formulates a plan based on the perceived state.

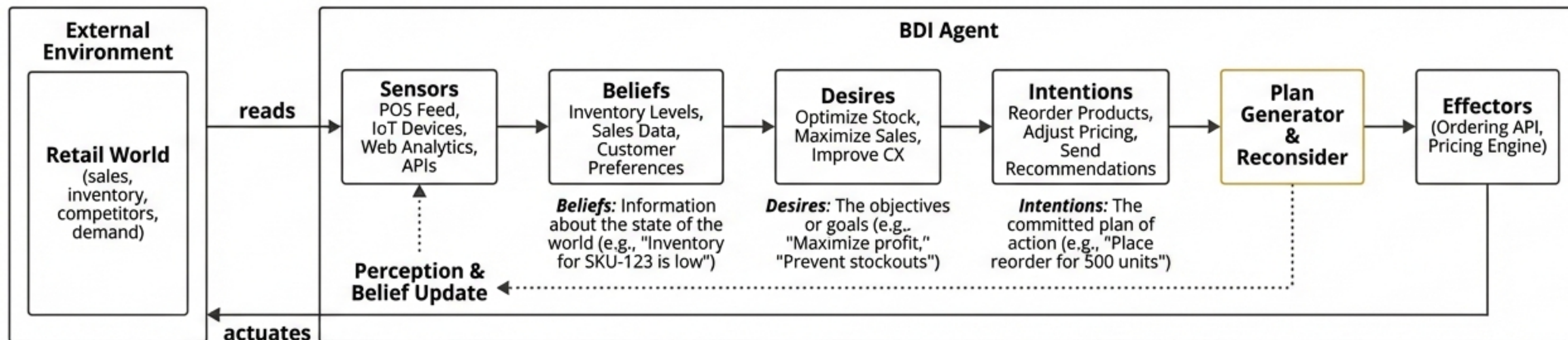
Act: Executes the plan via Effectors (APIs for price changes, Reorder protocols, Notifications).

Learn: Updates knowledge based on the outcome (Feedback mechanism), improving future performance.

Agent: A system capable of autonomous task performance, dynamic workflow design, and tool usage to achieve specific objectives.

Cognitive Architectures: How Agents Model the World

BDI Model (Belief-Desire-Intention)



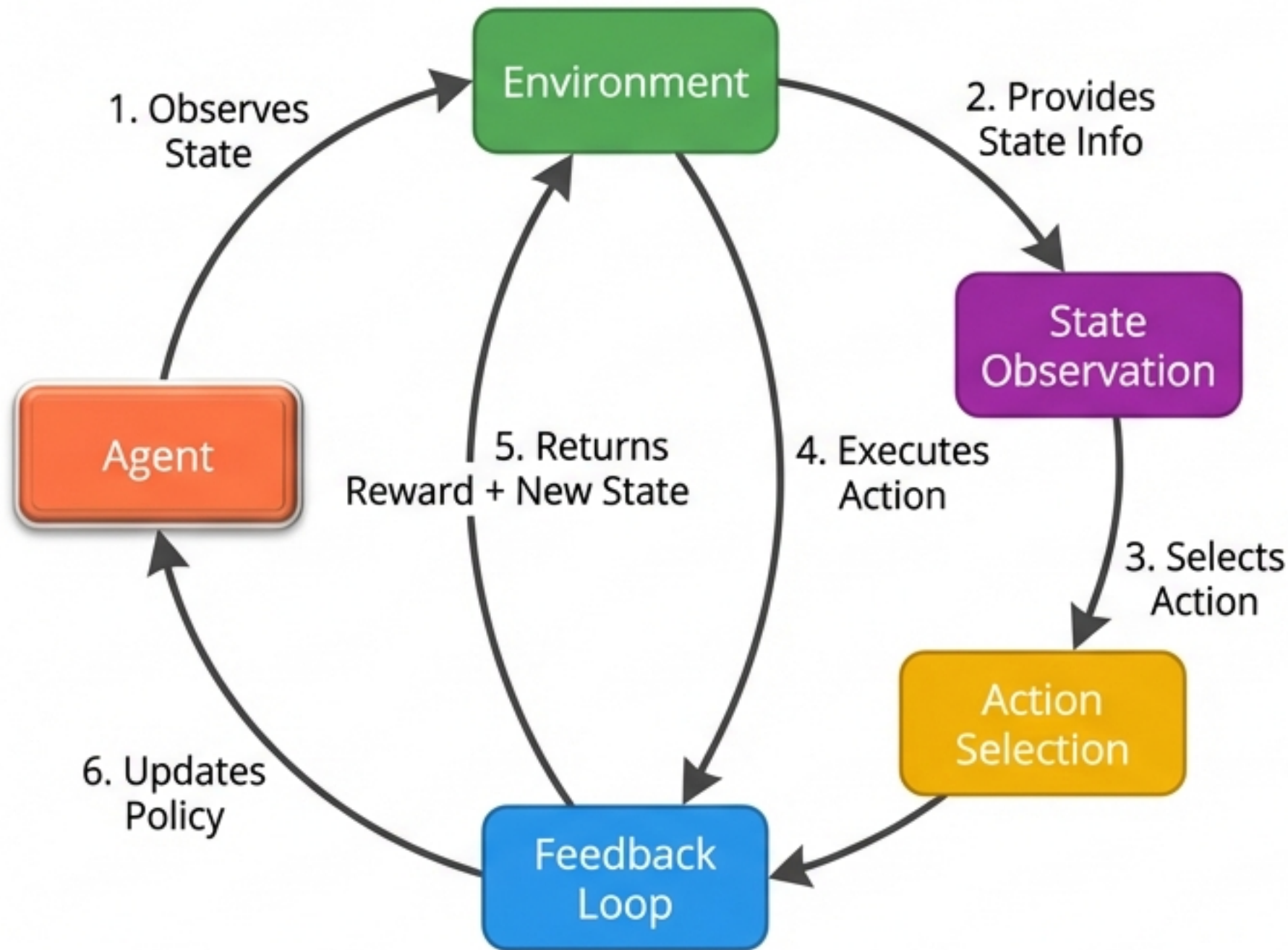
OODA Loop (Observe-Orient-Decide-Act)



A complementary framework emphasizing speed and agility. Critical for high-frequency trading or real-time dynamic pricing where market conditions shift in seconds.

Decision Frameworks: From Logic to Learning

Reinforcement Learning Cycle



Reinforcement Learning (RL):

Learning through interaction.

1. Agent observes the Environment (Store State).
2. Agent takes an Action (e.g., Markdown price).
3. Environment provides a Reward (e.g., Sales volume increase).
4. Agent updates its Policy to maximize future rewards.

Markov Decision Processes (MDPs):

Used for sequential decision-making where current actions affect future states. Essential for long-term planning where immediate profit must be balanced against future inventory health.

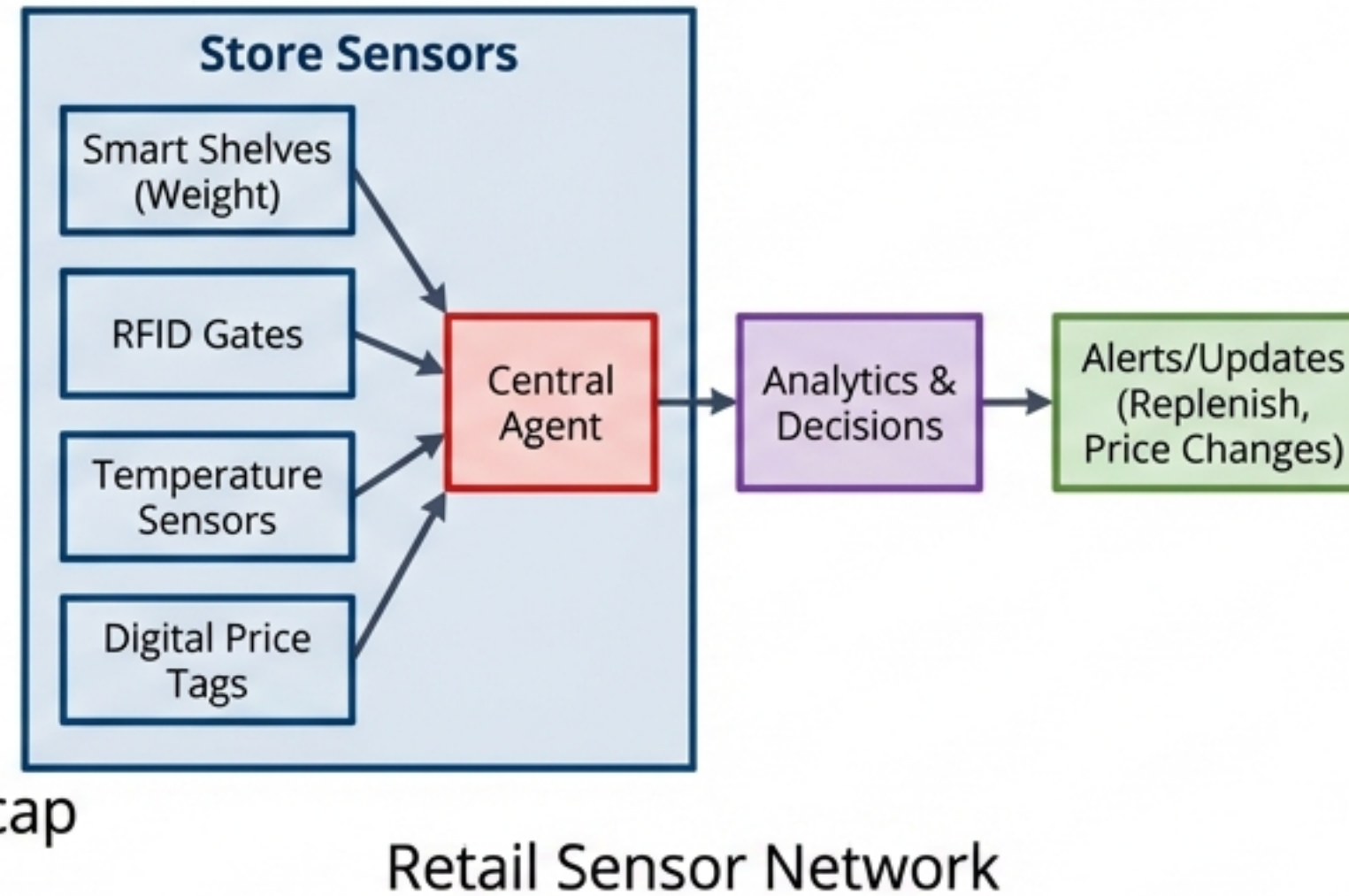
Use Case:

Dynamic Pricing—balancing immediate margin capture against long-term brand value.

Perception Layer: Giving Agents Eyes and Ears

Computer Vision (CV)

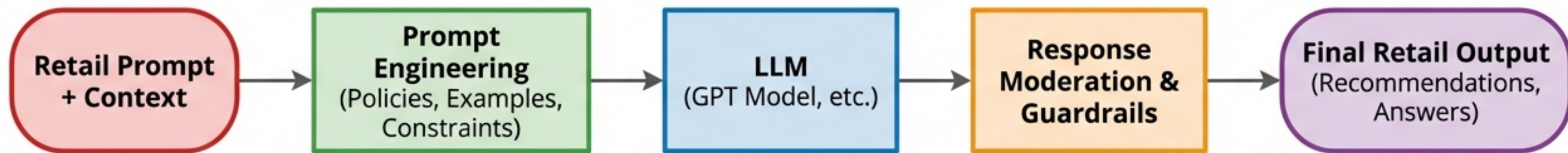
- **Planogram Compliance:**
Detecting if products are arranged correctly.
- **Visual Question Answering (VQA):**
Enabling staff to ask natural language questions like "Is the endcap fully stocked?" and receiving answers derived from camera feeds.



IoT & Sensors

- **Smart Shelves:**
Weight sensors detect product removal in real-time.
- **RFID/BLE:** Tracks asset location and customer journey mapping.
- **Environment:**
Temperature sensors for perishable goods monitoring.

Reasoning Layer: The Brain and The Context



LLM as Reasoning Engine

Large Language Models (LLMs)



The cognitive engine.

Handles planning, reasoning, and natural language interfaces.

Translates high-level goals ("Clear summer stock") into specific API calls.

Knowledge Graphs (KGs)



The "Context."

Maps semantic relationships (e.g., Product A is compatible with Product B).

Enables complex queries: "Show me substitutes for this out-of-stock item."

Causal Reasoning



Understanding *why*.

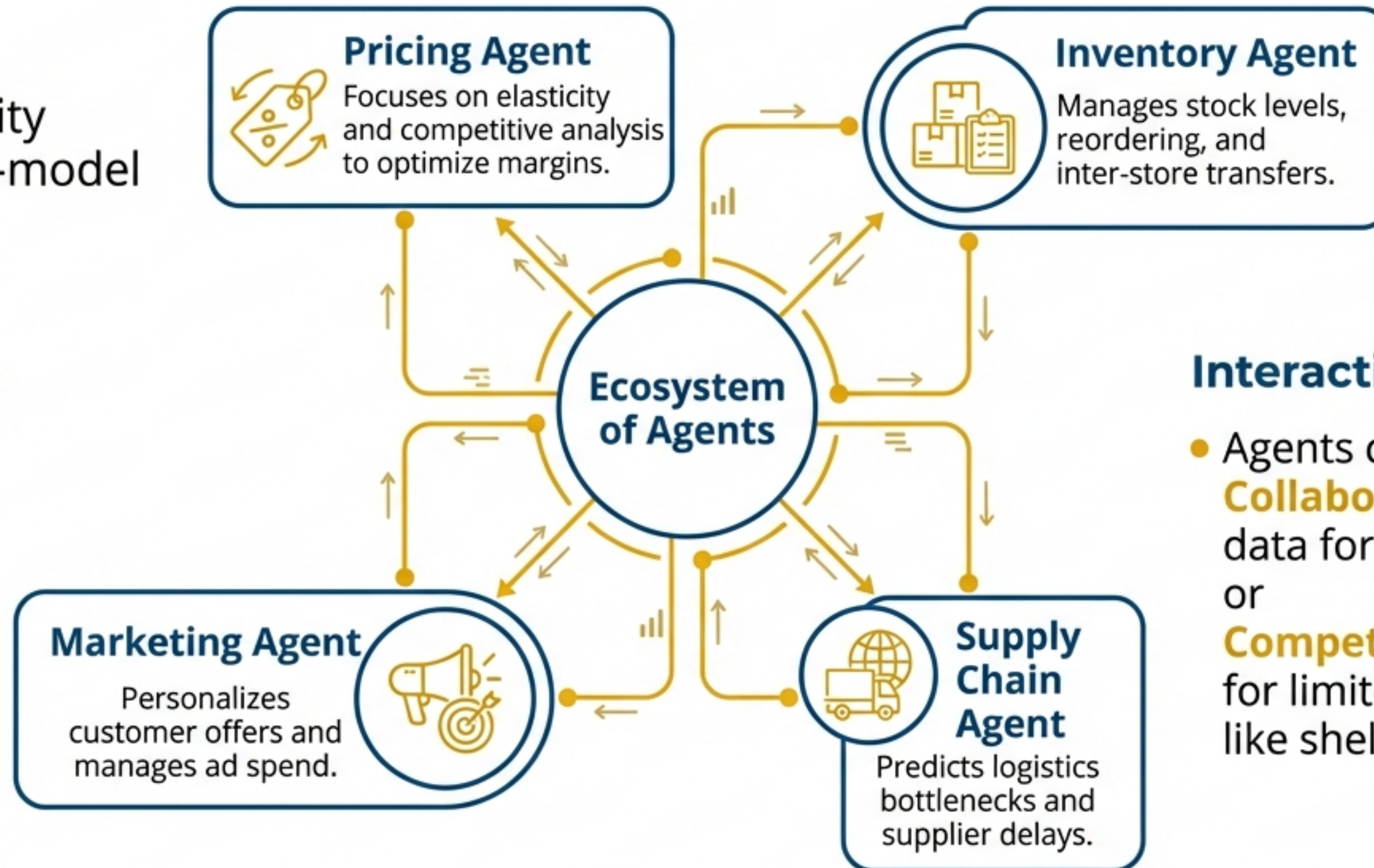
Moves beyond correlation to causation (e.g., "Did sales drop because of the price increase, or because it was raining?").

Multi-Agent Systems (MAS): The Specialized Workforce

Why MAS?

Retail complexity exceeds single-model capabilities.

MAS offers Specialization, Scalability, and Resilience.



Interaction Dynamics:

- Agents can be **Collaborative** (pooling data for a shared goal) or **Competitive** (bidding for limited resources like shelf space).

Coordination and Negotiation Protocols



FIPA Message Example

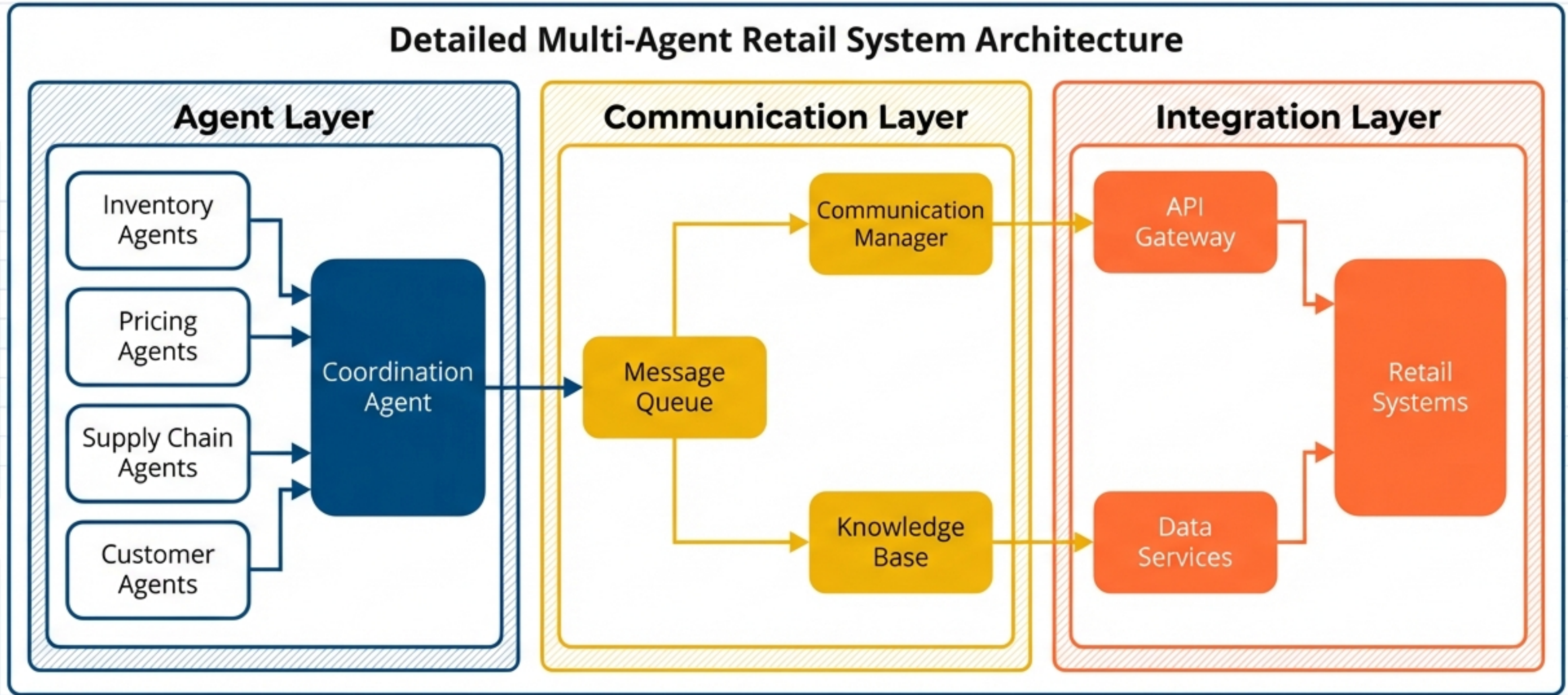
Communication Standards:

- **FIPA (Foundation for Intelligent Physical Agents):** Standardized message formats for agent interoperability.
- **A2A (Agent-to-Agent):** Emerging protocols allowing agents from different vendors to collaborate.

Coordination Mechanisms:

- **Negotiation:** Agents bargain to resolve conflicts. *Example:* Marketing Agent wants a promo; Inventory Agent warns of low stock -> They negotiate a limited-time offer.
- **Auctions:** Market mechanisms to allocate scarce resources. *Example:* Brands (via agents) bidding for premium endcap display space.

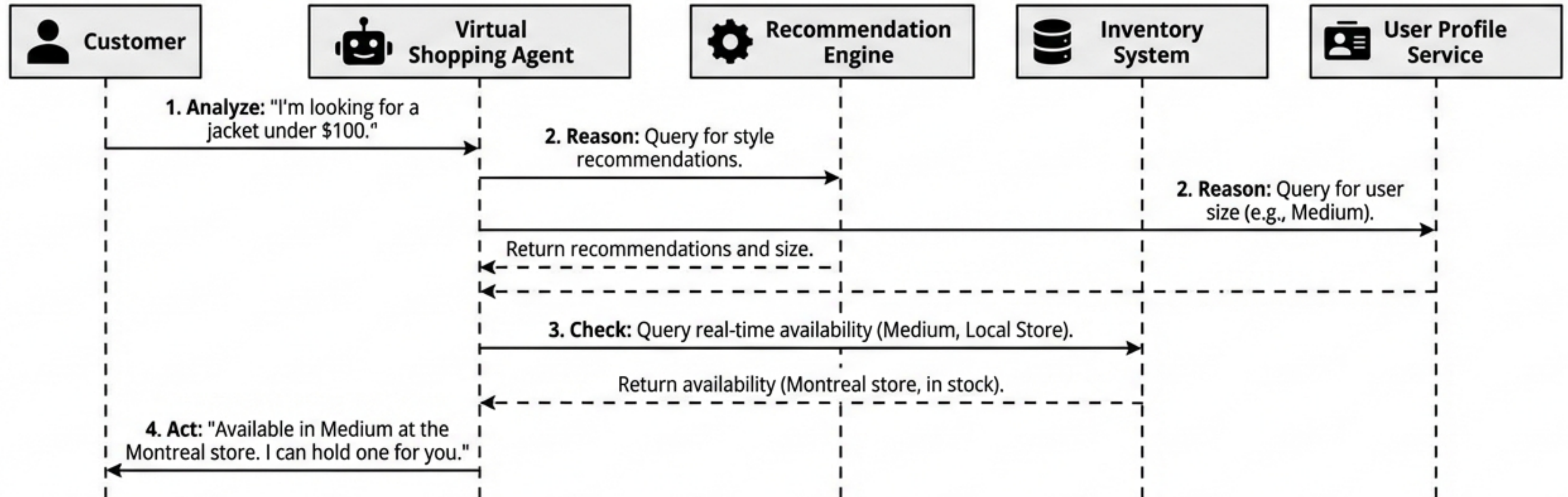
Reference Architecture: The 3-Layer Blueprint



A scalable architecture separating agent logic, communication orchestration, and legacy integration.

End-to-End Integration in Action

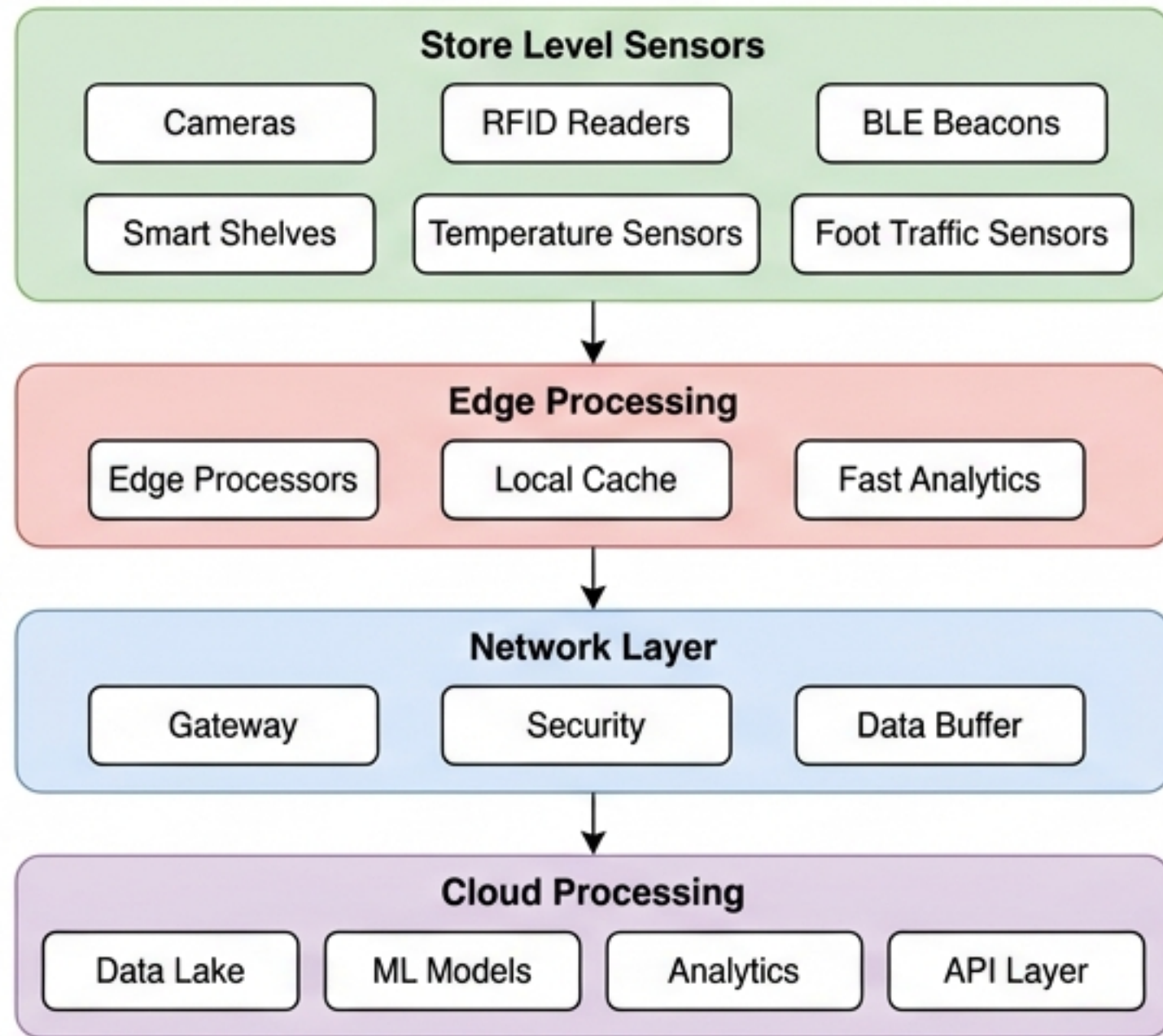
The Scenario: Virtual Shopping Agent finds a jacket under \$100.



Key Insight: This requires real-time, event-driven integration across siloed domains (CX, Merchandising, Supply Chain).

Implementation Strategy: Cloud vs. Edge

Sensor Network Architecture



Edge Computing (In-Store):

Runs low-latency agents (CV, robotics). Ensures real-time response and data privacy (video feeds processed locally).

Cloud Computing (Central):

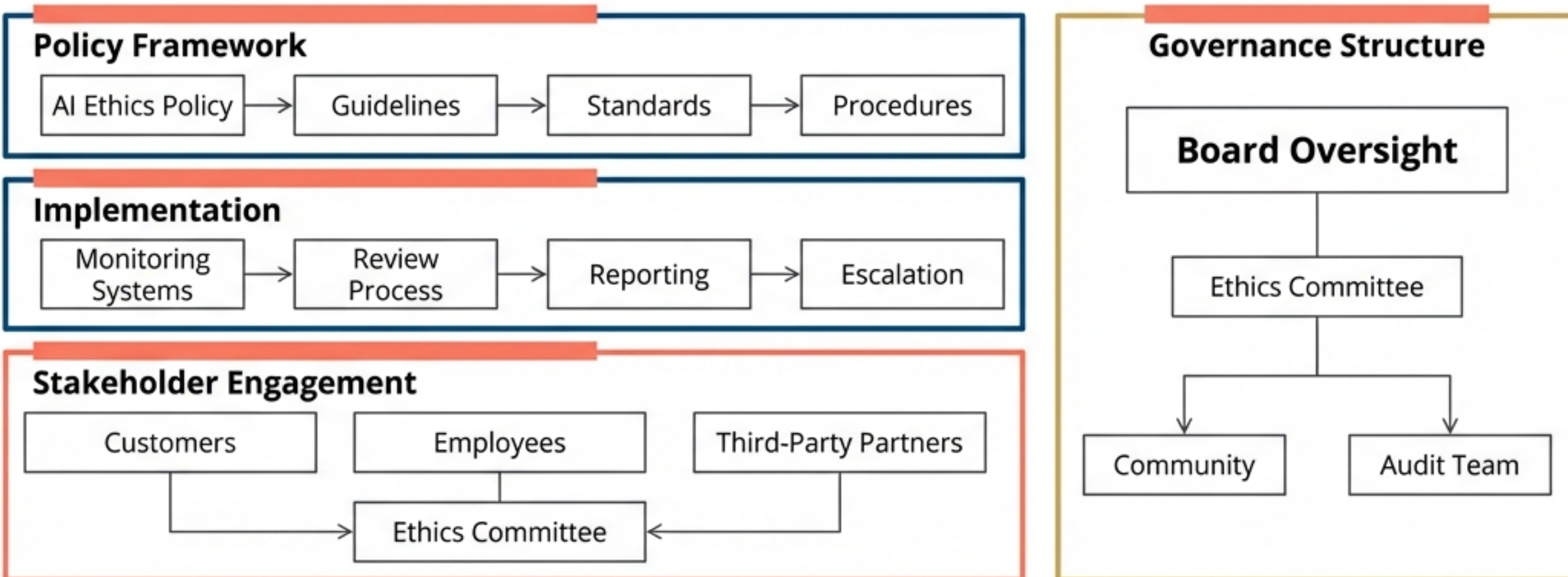
Handles heavy lifting (LLM training, global aggregation, long-term analytics).

Hybrid Architecture:

The standard for modern retail, balancing speed (Edge) with intelligence depth (Cloud). (Cloud).

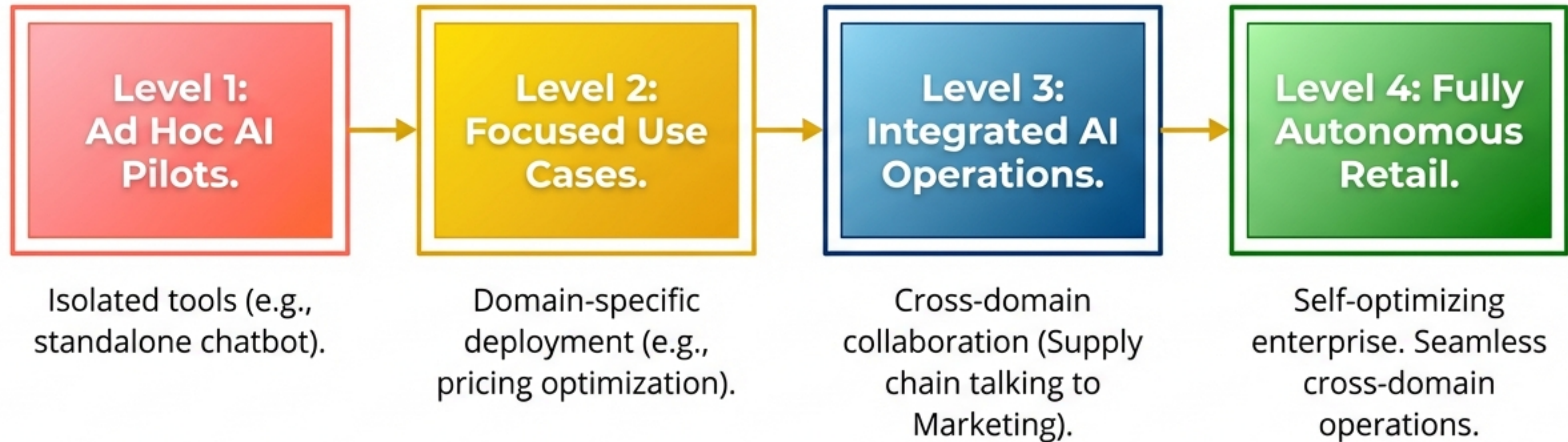
Governance: Guardrails and Human-in-the-Loop

Ethical Governance Framework



- **Human-in-the-Loop (HITL):** High-stakes decisions (e.g., >20% price drop) require human approval.
- **Transparency:** Use **Model Cards** to document agent limitations and training data.
- **Explainability:** Interfaces must explain *why* a decision was made.
- **Safety:** Strict guardrails against bias and hallucination.

The Maturity Model: A Roadmap to Autonomy



The Goal: Humans shift from routine operational tasks to strategic oversight and creative innovation.



Agentic AI converges autonomy, data, and strategy to create retail systems that don't just predict the future, but actively shape it.

Start with modular, specialized agents.
Ensure strong governance.
Scale toward the fully autonomous enterprise.

Content based on "Foundations of Agentic AI for Retail" (2025) by Dr. Fatih Nayebi. Presentation designed by Graph4VA.